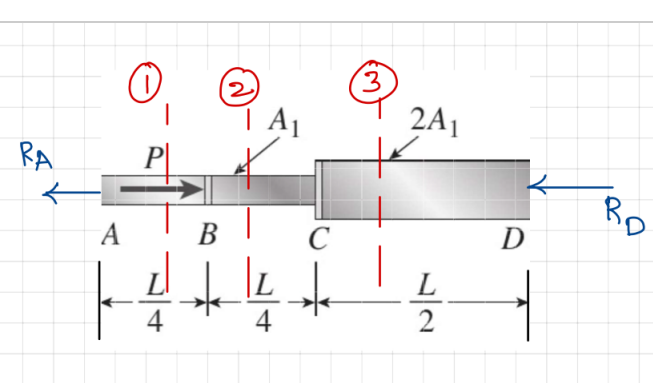
Tutorial-5 Solutions

Wednesday, 23 September 2026 10:45 PM

1)



Section ①

$N_1 = R_A$

Section ②

$N_2 = R_A - P$

Section ③

$N_3 = -R_D$

Equation of Equillibrium

$$\xrightarrow{+} \Sigma F_x = 0 \Rightarrow -R_A + P - R_D = 0$$

$$R_A + R_D = P \quad \text{——} ①$$

Equation of compatibility

$$\delta_{AB} + \delta_{BC} + \delta_{CD} = 0$$

$$\frac{R_A(L/4)}{EA_1} + \frac{(R_A-P)(L/4)}{EA_1} + \frac{(-R_D)(L/2)}{2EA_1} = 0$$

$$R_A + R_A - P - R_D = 0$$

$$2R_A - R_D = P \quad \text{——} ②$$

(a) Reactions

Solving ① & ②

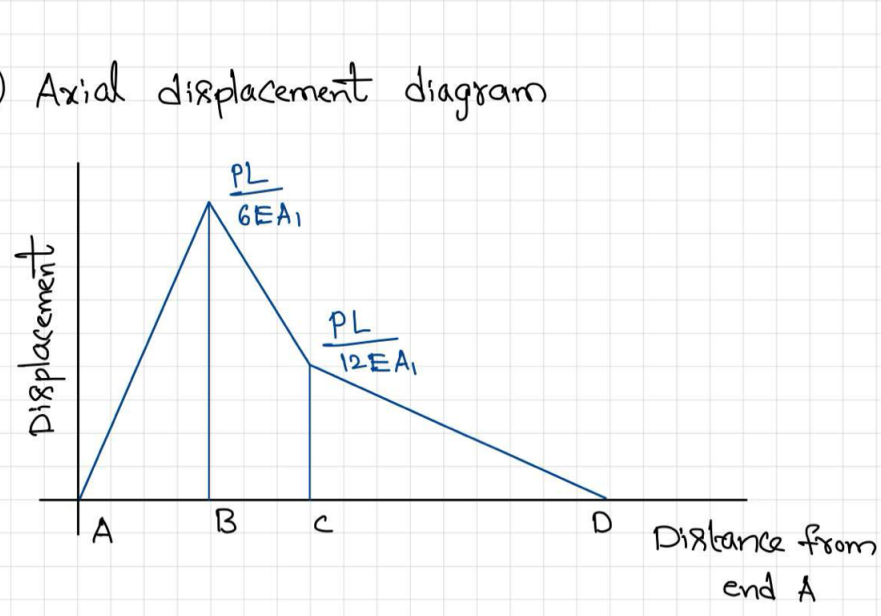
$$R_A = \frac{2P}{3}, \quad R_D = P/3$$

(b) Displacements at points B and C

$$\delta_B = \delta_{AB} = \frac{R_A L}{4EA_1} = \frac{PL}{6EA_1} \quad (\text{To the right})$$

$$\delta_C = \delta_{AB} + \delta_{BC} = \frac{PL}{6EA_1} - \frac{PL}{12EA_1} = \frac{PL}{12EA_1} \quad (\text{to the right})$$

(c) Axial displacement diagram



Tutorial - 5 Solutions

Saturday, 19 September 2026 4:29 PM

Question - 1 (Energy Method)

Bar is held at both ends A & D.

We divide bar into 3 segments:

AB, BC & CD.

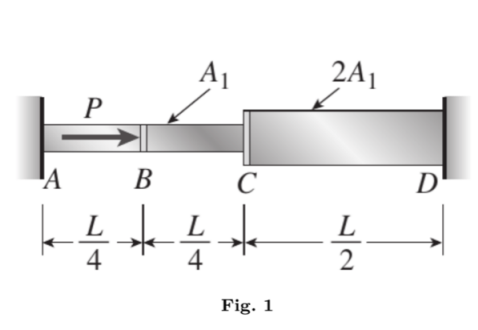


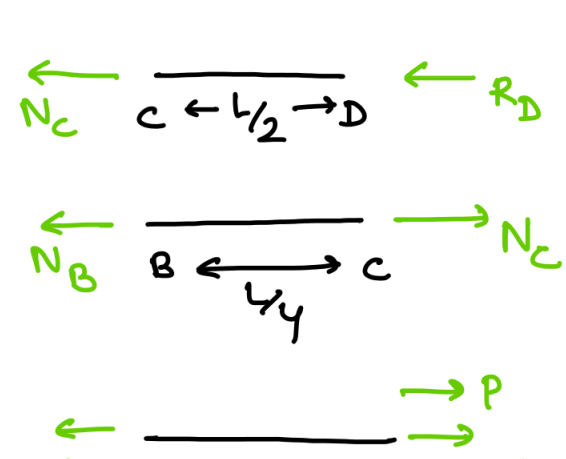
Fig. 1

Now, using equilibrium condition :-

$N_C = -R_D$ (CD)

$N_B = N_C = -R_D$ (BC)

$N_A = P + N_B = P - R_D$



Now, strain energy for an axially loaded bar is given by →

$$U = \sum \frac{N^2 L}{2AE}$$

$$U = U_{AB} + U_{BC} + U_{CD}$$

$$U = \frac{(P-R_D)^2(L/4)}{2A_1E} + \frac{(-R_D)^2(L/4)}{2A_1E} + \frac{(-R_D)^2(L/2)}{2(2A_1)E}$$

$$U = \frac{L}{8A_1E}(P-R_D)^2 + \frac{R_D^2 L}{A_1E}\left[\frac{L}{8} + \frac{L}{8}\right]$$

$$U = \frac{L}{8A_1E}\left[(P-R_D)^2 + 2R_D^2\right]$$

According to Castigliano's theorem →

$$\frac{\partial U}{\partial R_D} = 0$$

$$\therefore \frac{L}{8A_1E}\left[-2(P-R_D) + 4R_D\right] = 0$$

$$\Rightarrow -2P + 2R_D + 4R_D = 0$$

$$\Rightarrow \boxed{R_D = P/3}$$

$$R_A + R_D = P \Rightarrow R_A = P - P/3$$

$$\boxed{R_A = \frac{2P}{3}}$$

(b) To determine δ_B, we take partial derivate of U w.r.t P because at B, load P is acting.

$$\therefore \delta_B = \frac{\partial}{\partial P}\left[\frac{L}{8A_1E}\left[(P-R_D)^2 + 2R_D^2\right]\right]$$

$$\delta_B = \frac{L}{8A_1E}\left[2(P-R_D) + 0\right]$$

$$\delta_B = \frac{2L(P-R_D)}{8A_1E}$$

$$\because R_D = P/3 \Rightarrow \delta_B = \frac{L}{4A_1E}\left[P - P/3\right]$$

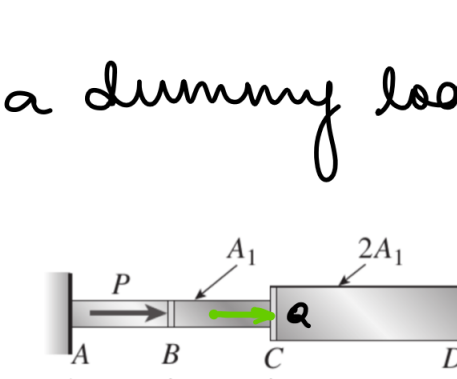
$$\boxed{\delta_B = \frac{PL}{6A_1E}}$$

To determine C, we need to give a dummy load 'Q' at C. :-

$N_{CD} = -R_D$

$N_{BC} = Q - R_D$

$N_{AB} = P + Q - R_D$



New strain energy (U') →

$$U' = \frac{L}{8A_1E}\left[(P+Q-R_D)^2 + (Q-R_D)^2 + R_D^2\right]$$

For compatibility condition for R_D →

$$\frac{\partial U'}{\partial R_D} = 0$$

$$\Rightarrow \frac{L}{8A_1E}\left[-2(P+Q-R_D) + (-2)(Q-R_D) + 2R_D\right] = 0$$

$$\Rightarrow -2(P+Q) + 2R_D - 2Q + 2R_D + 2R_D = 0$$

$$\Rightarrow 6R_D = 2P + 2Q + 2Q$$

$$\Rightarrow R_D = \frac{P+2Q}{3}$$

Now, $\delta_C = \frac{\partial U'}{\partial Q}$

$$\delta_C = \frac{L}{8A_1E}\left[2(P+Q-R_D) + 2(Q-R_D)\right]$$

$$\delta_C = \frac{L}{8A_1E}\left[2P + 2Q - 2R_D + 2Q - 2R_D\right]$$

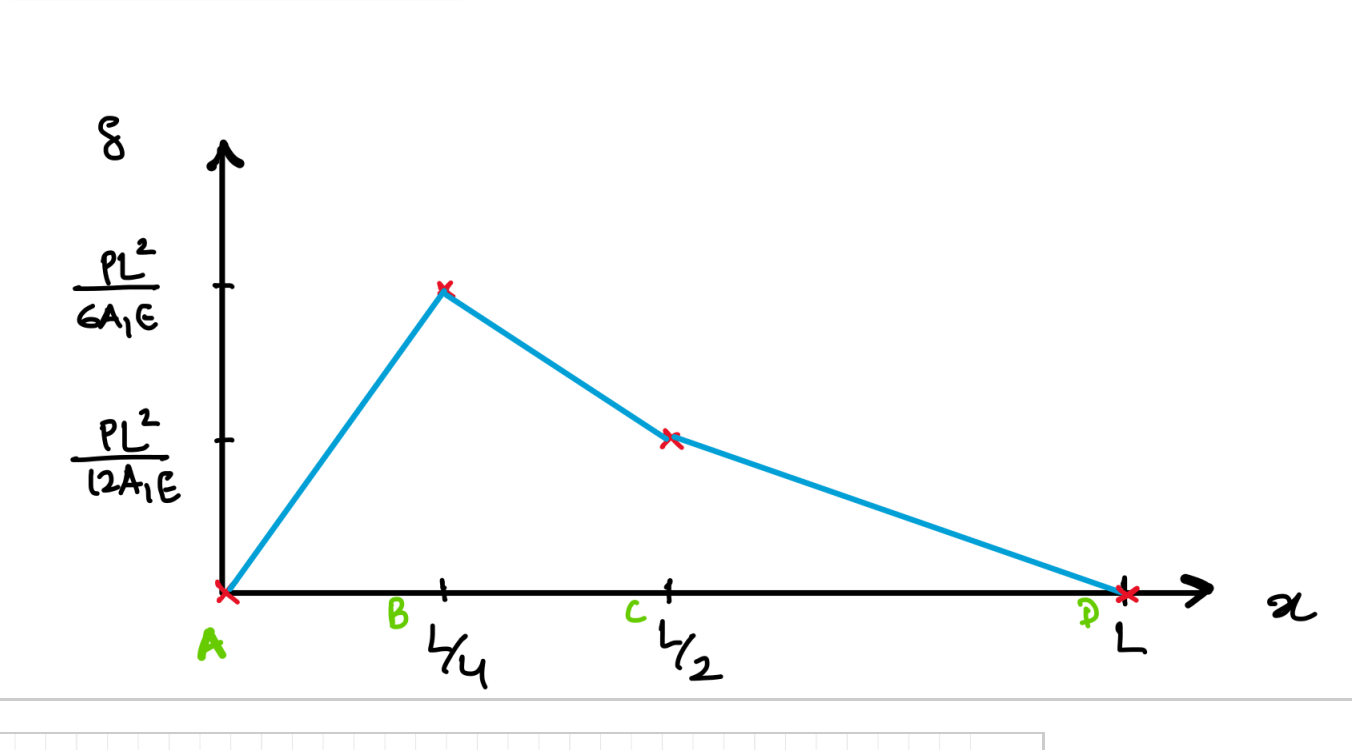
$$\delta_C = \frac{L}{4A_1E}\left[P + 2Q - 2\left[\frac{P+2Q}{3}\right]\right]$$

As Q is dummy load, we substitute Q = 0 →

$$\delta_C = \frac{L}{4A_1E}\left[P - \frac{2P}{3}\right]$$

$$\boxed{\delta_C = \frac{PL}{12A_1E}}$$

(c)



AM2200 Tutorial 5 solutions

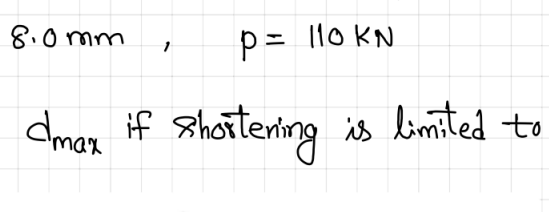
2) Numerical data:-

$d_1 = 100$ mm, $d_2 = 60$ mm

$L = 1200$ mm, $E = 4.0$ Gpa

$\delta_a = 8.0$ mm, $P = 110$ kN

(a) find d_{max} if shortening is limited to $\delta_a = 8$ mm



$$A_1 = \frac{\pi}{4}d_1^2, \quad A_2 = \frac{\pi}{4}d_2^2$$

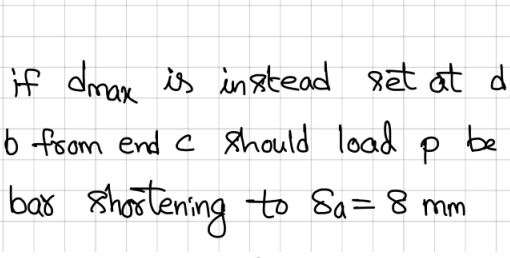
$$\delta_a = \frac{P(L/4)}{E\frac{\pi}{4}(d_1^2 - d_{max}^2)} + \frac{P(L/4)}{EA_1} + \frac{P(L/2)}{EA_2}$$

$$\delta_a = \frac{PL}{E\pi(d_1^2 - d_{max}^2)} + \frac{PL}{E\pi d_1^2} + \frac{PL}{E\pi d_2^2}$$

$$\frac{E\pi\delta_a}{PL} = \frac{1}{(d_1^2 - d_{max}^2)} + \frac{1}{d_1^2} + \frac{2}{d_2^2}$$

$$\underline{d_{max}} = 23.9\text{mm}$$

(b) Now, if d_{max} is instead set at $d_2/2$, at what distance b from end C should load P be applied to limit the bar shortening to $\delta_a = 8$ mm


$d_{max} = \frac{d_2}{2}$
A B C
d_2
d_1
P P
$\frac{L}{4}$ $\frac{L}{4}$ $\frac{L}{2}$
b

$$A_0 = \frac{\pi}{4}\left(d_1^2 - \left(\frac{d_2}{2}\right)^2\right)$$

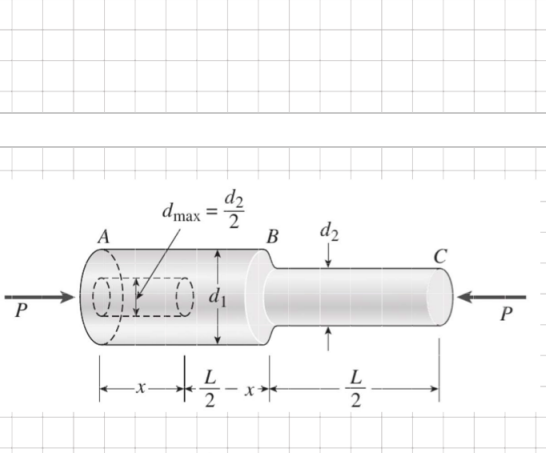
$$\delta_a = \frac{P(L/4)}{EA_0} + \frac{P(L/4)}{EA_1} + \frac{P(L/2 - b)}{EA_2}$$

$$\delta_a = \frac{PL}{E\pi\left(d_1^2 - \left(\frac{d_2}{2}\right)^2\right)} + \frac{PL}{E\pi d_1^2} + \frac{2P(L-2b)}{E\pi d_2^2}$$

$$\frac{E\pi\delta_a}{P} = \frac{L}{\left(d_1^2 - \left(\frac{d_2}{2}\right)^2\right)} + \frac{L}{d_1^2} + \frac{2(L-2b)}{d_2^2}$$

$$\underline{b} = 4.16\text{ mm}$$

(c) finally if loads P are applied at the ends and $d_{max} = d_2/2$, what is the permissible length x of the hole if shortening is to be limited to $\delta_a = 8$mm.



$$\delta = \frac{Px}{EA_0} + \frac{P\left(\frac{L}{2} - x\right)}{EA_1} + \frac{P(L/2)}{EA_2}$$

$$\frac{\delta E}{P} = \frac{x}{A_0} - \frac{x}{A_1} + \frac{L}{2}\left(\frac{1}{A_1} + \frac{1}{A_2}\right)$$

$$x\left(\frac{1}{A_0} - \frac{1}{A_1}\right) = \frac{\delta E}{P} - \frac{L}{2}\left(\frac{1}{A_1} + \frac{1}{A_2}\right)$$

$$x\left(\frac{A_1 - A_0}{A_0 A_1}\right) = \frac{\delta E}{P} - \frac{L}{2}\left(\frac{1}{A_1} + \frac{1}{A_2}\right)$$

$$x = \frac{A_0 A_1\left[\frac{\delta E}{P} - \frac{L}{2}\left(\frac{1}{A_1} + \frac{1}{A_2}\right)\right]}{A_1 - A_0}$$

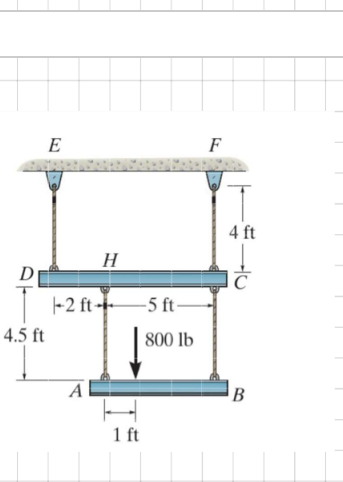
$A_0 = 7.147123 \times 10^{-3}$ m²

$A_1 = 7.853981 \times 10^{-3}$ m²

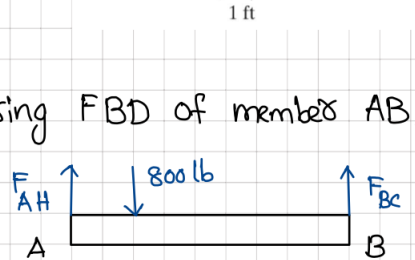
$A_2 = 2.827433 \times 10^{-3}$ m²

$$\underline{x} = 183.3 \text{ mm}$$

3)



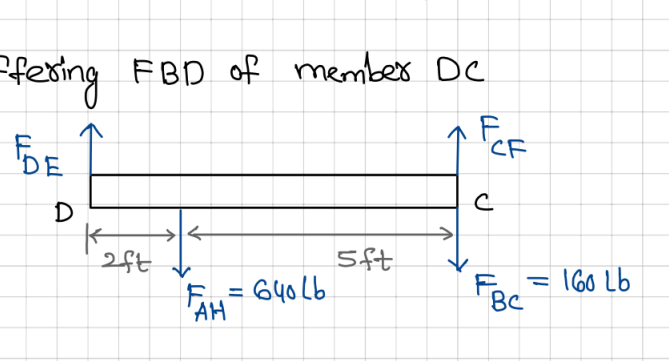
Reffering FBD of member AB,



$$\circlearrowleft + \Sigma M_A = 0 \Rightarrow F_{BC}(5) - 800(1) = 0 \Rightarrow F_{BC} = 160 \text{ lb}$$

$$\circlearrowleft + \Sigma M_B = 0 \Rightarrow 800(4) - F_{AH}(5) = 0 \Rightarrow F_{AH} = 640 \text{ lb}$$

Reffering FBD of members DC



$$\circlearrowleft + \Sigma M_D = 0 \Rightarrow F_{CF}(7) - 160(7) - 640(2) = 0$$

$$\Rightarrow F_{CF} = 342.86\text{lb}$$

$$\circlearrowleft + \Sigma M_C = 0 \Rightarrow 640(5) - F_{DE}(7) = 0$$

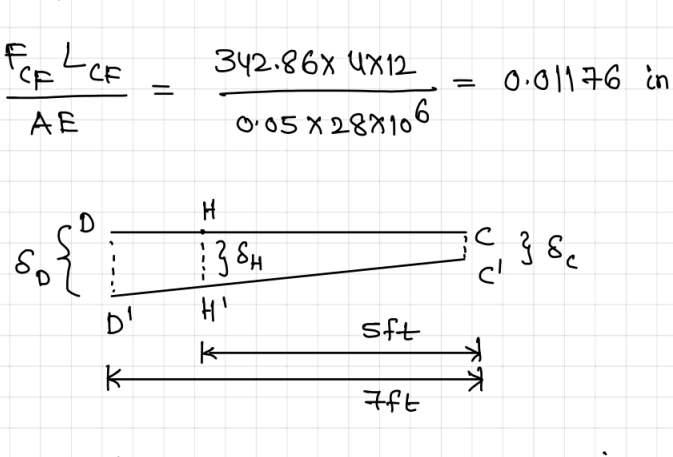
$$\Rightarrow F_{DE} = 457.14\text{lb}$$

Since E and F are fixed

$$\delta_D = \frac{F_{DE}L_{DE}}{AE} = \frac{457.14 \times 4 \times 12}{0.05 \times 28 \times 10^6} = 0.01567 \text{ in} \downarrow$$

$$\delta_C = \frac{F_{CF}L_{CF}}{AE} = \frac{342.86 \times 4 \times 12}{0.05 \times 28 \times 10^6} = 0.01176 \text{ in} \downarrow$$

From geometry



$$\delta_H = 0.01176 + \frac{5}{7}(0.01567 - 0.01176) = 0.01455 \text{ in} \downarrow$$

Subsequently

$$\delta_{A/H} = \frac{F_{AH}L_{AH}}{AE} = \frac{640 \times 4.5 \times 12}{0.05 \times 28 \times 10^6} = 0.02469 \text{ in} \downarrow$$

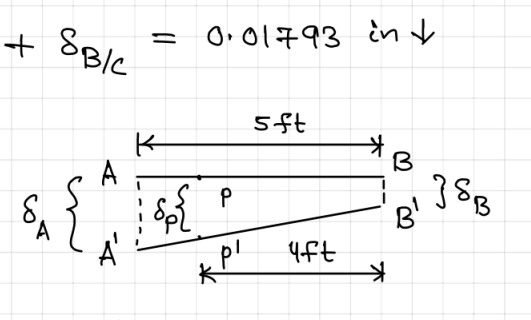
$$\delta_{B/C} = \frac{F_{BC}L_{BC}}{AE} = \frac{160 \times 4.5 \times 12}{0.05 \times 28 \times 10^6} = 0.006171 \text{ in} \downarrow$$

thus

$$\delta_A = \delta_H + \delta_{A/H} = 0.03924 \text{ in} \downarrow$$

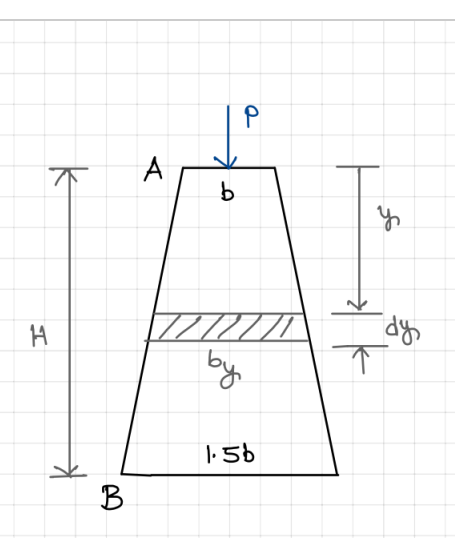
$$\delta_B = \delta_C + \delta_{B/C} = 0.01793 \text{ in} \downarrow$$

From geometry



$$\underline{\delta_P} = 0.01793 + \frac{4}{5}(0.03924 - 0.01793) = \underline{0.0350} \text{ in} \downarrow$$

4)



width at A = b

width at B = 1.5b

width at distance y = b_y

$$b_y = b + (1.5b - b)\frac{y}{H}$$

$$= \frac{b}{H}(H + 0.5y)$$

cross sectional area at distance y = $A_y = (b_y)^2$

$$A_y = \frac{b^2}{H^2}(H + 0.5y)^2$$

Shortening of element dy

$$d\delta = \frac{P\,dy}{EA_y} = \frac{P\,dy}{E\frac{b^2}{H^2}(H+0.5y)^2}$$

shortening of entire part

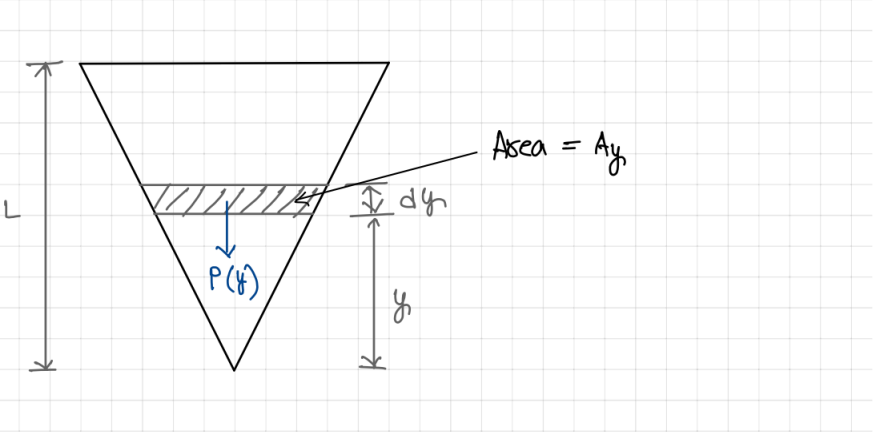
$$\delta = \int d\delta = \frac{PH^2}{Eb^2}\int_0^H \frac{dy}{(H+0.5y)^2}$$

$$\left[\int \frac{dx}{(a+bx)^2} = -\frac{1}{b(a+bx)}\right]$$

$$\delta = \frac{PH^2}{Eb^2}\left[-\frac{1}{(0.5)(H+0.5y)}\right]_0^H$$

$$\underline{\delta} = \frac{2PH}{3Eb^2}$$

5)



$$P(y) = \frac{1}{3}\gamma A_y y$$

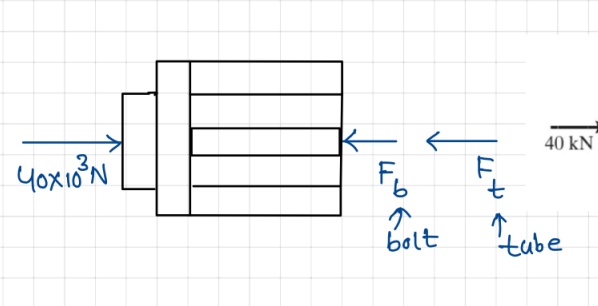
$$d\delta = \frac{P(y)\,dy}{EA_y}$$

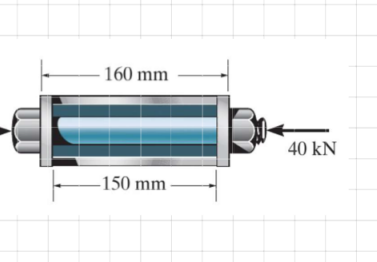
$$\delta = \int_0^L d\delta = \int_0^L \frac{P(y)\,dy}{EA_y}$$

$$= \int_0^L \frac{\frac{1}{3}\gamma A_y y\,dy}{EA_y} = \frac{\gamma}{3E}\int_0^L y\,dy$$

$$\underline{\delta} = \frac{\gamma L^2}{6E}$$

6)





Reffering to the FBD of the left portion of the cut assembly,

$$\xrightarrow{+} \Sigma F_x = 0 \Rightarrow 40 \times 10^3 - F_b - F_t = 0$$

$$F_b + F_t = 40 \times 10^3 \quad \text{——} ①$$

The bolt and tube have the same deformation. Thus

$$\delta_t = \delta_b$$

$$\frac{F_t(150)}{\frac{\pi}{4}(0.06^2 - 0.05^2)(200 \times 10^9)} = \frac{F_b(160)}{\frac{\pi}{4}(0.02)^2(200 \times 10^9)}$$

$$F_t = 2.9333 F_b \quad \text{——} ②$$

Solving ① & ② yields

$$F_b = 10.17 \times 10^3 \text{N}$$

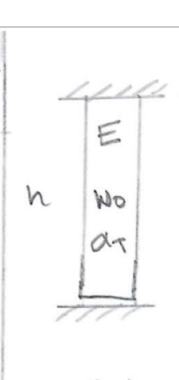
$$F_t = 29.83 \times 10^3 \text{N}$$

Thus,

$$\sigma_b = \frac{F_b}{A_b} = \frac{10.17 \times 10^3}{\frac{\pi}{4} \times 0.02^2} = 32.4 \text{ MPa}$$

$$\sigma_t = \frac{F_t}{A_t} = \frac{29.83 \times 10^3}{\frac{\pi}{4}(0.06^2 - 0.05^2)} = 34.5 \text{ MPa}$$

7.



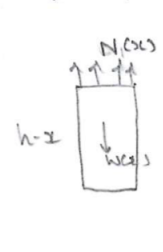
a) as the bar just touches the ground once heated strain at x=h will be immediately zero

Let σ_2 be the stress of support reaction from bottom

at h, $\varepsilon = \frac{\sigma_2}{E} + \alpha_T \Delta T = 0$

$\Rightarrow \sigma_2 = -E\alpha_T \Delta T$

Let $\sigma_1(x)$ be stress only due to self weight Wo



$N(x) = Wo\frac{(h-x)}{h} = Wo\left(1 - \frac{x}{h}\right)$ (weight of lower portion of bar)

$\therefore \sigma_1 = \frac{N(x)}{A} = \frac{Wo}{A}\left(1 - \frac{x}{h}\right)$

∴ Total stress $\sigma(x) = \sigma_1 + \sigma_2$

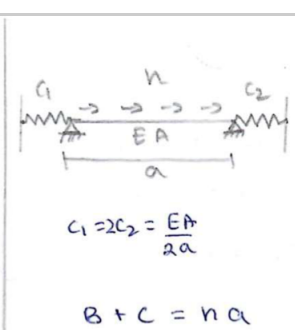
$= \frac{Wo}{A}\left(1 - \frac{x}{h}\right) - E\alpha_T \Delta T$

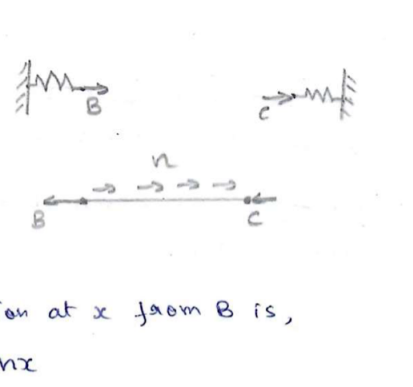
b) due to blocking of temperature strain, the end of the bar is always under compression. As, stress distribution is linear stress will be compressible every where it compression is present at the top.

$\sigma(x=0) < 0 \Rightarrow \frac{Wo}{A} - E\alpha_T \Delta T < 0$

$\Rightarrow \Delta T > \frac{Wo}{EA\alpha_T}$

8.





$c_1 = 2c_2 = \frac{EA}{2a}$

$B + C = na$

Normal force for a section at x from B is,

$N(x) = B - nx$

Elongation at B & C is,

$\Delta u_B = \frac{B}{c_1} = \frac{B(2a)}{EA}$

$\Delta u_C = \frac{C}{c_2} = \frac{C(4a)}{EA}$

Strain in bar $\varepsilon = \frac{N(x)}{EA} = \frac{B - nx}{EA}$

$\frac{du}{dx} = \frac{B - nx}{EA} \Rightarrow \Delta u_{bar} = \int_0^a \frac{B - nx}{EA} = \frac{Ba}{EA} - \frac{na^2}{2EA}$

$\Delta u_1 + \Delta u_{bar} = \Delta u_2$

$\Rightarrow \frac{2Ba}{EA} + \frac{Ba}{EA} - \frac{na^2}{2EA} = \frac{4Ca}{EA} = \frac{4(na - B)a}{EA}$

$\Rightarrow \frac{7Ba}{EA} = \frac{9na^2}{2EA} \Rightarrow B = \frac{9}{14}na$

$\Rightarrow N(x) = \frac{9na}{14} - nx$